Unit 2 – Selection and Iteration Name: **KEY**
Checkpoint Practice 1

1) Label the following as true (T) or false (F).

**False** a) A Boolean type variable can store a `true`, `false`, or `null` value. (cannot be null)

**True** b) The relational operators will evaluate to a Boolean value.

**True** c) In the expression `x && y`, short-circuit means that if x evaluates to false, the expression y will not be checked.

**False** d) In the expression `x || y`, short-circuit means that if x evaluates to false, the expression y will not be checked. if x evaluates to true, the y expression will not be checked

**True** e) In order precedence, relational operators get evaluated before Boolean operators

**False** f) In order precedence, compound operators get evaluated before Boolean operators compound operators happen last!

**False** g) `!(X || Y)` is equivalent to `!X || !Y` it is equivalent to `!X && !Y`

**True** h) `!(X && Y)` is equivalent to `!X || !Y`

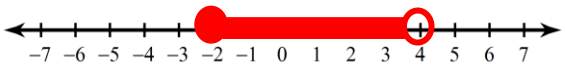
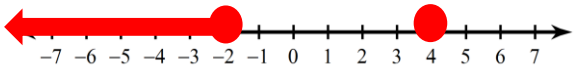
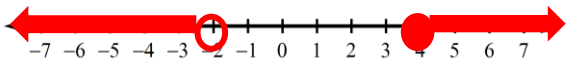
2) Translate each of the following English statements into logical tests that could be used in an if/else statement. Assume that three int variables, x, y, and z, have already been declared.

| | |
|---|---|
| a) z is odd. | `z % 2 == 1` |
| b) z is not greater than y's square root. | `z <= Math.sqrt(y)` |
| c) y is positive. | `y > 0` |
| d) Either x or y is even. | `x % 2 == 0 || y % 2 == 0` |
| e) y is a multiple of z. | `y % z == 0` |
| f) z is not zero. | `z != 0` |
| g) x and z are of opposite signs. | `x * z < 0` |
| h) y is a nonnegative one-digit number. | `y >= 0 && y <= 9` |

3) Match the equivalent statements on the right to the ones on the left.

| | | |
|---|---|---|
| C | `!(grade > 90)` | A) `!(grade < 90)` |
| B | `grade > 90` | B) `!(grade <= 90)` |
| D | `!(grade == 90)` | C) `grade <= 90` |
| A | `grade >= 90` | D) `grade != 90` |

4) Match the algebraic expression or number line with the Java code. Assume `x` is an integer value.

| | | |
|---|---|---|
| B |  | A) `x < -2 \|\| x > 4` |
| E | $3 < x < 5$ | B) `x >= -2 && x < 4` |
| C |  | C) `x <= -2 \|\| x == 4` |
| D | $-2 < x \leq 4$ | D) `x > -2 && x <= 4` |
| F | $x \leq 3$ or $x \geq 5$ | E) `x == 4` |
| A |  | F) `x != 4` |

5) `A` and `B` are boolean values. Complete the truth table below.

| A | B | !A | !A \|\| B | A && !B | (A && !B) \|\| !A |
|---|---|---|---|---|---|
| T | T | F | T | F | F |
| T | F | F | T | T | T |
| F | T | T | T | F | T |
| F | F | T | T | F | T |

6) You want to develop a program to play the lottery. The program randomly generates a lottery of a two-digit number, prompts the user the enter the two-digit number (10 – 99), and then determines if the user wins according to the following rules:

a) If the user matches the lottery number exactly, the award is $10,000
b) If the user matches the lottery number within ±4, the award is $3,000
c) If the user matches the lottery number with ±8, the award is $1,000

Complete the following code to write the program described above.

```
import java.util.Scanner;

public class Lottery
{
	public static void main(String[] args)
	{
		int lotteryNum = (int)(Math.random() * 90) + 10;

		Scanner keyboard = new Scanner(System.in);
		System.out.println("Enter your two-digit lottery pick: ");
		int guess = keyboard.nextInt();

		//Check the values
		if(lotteryNum == guess)
		{
			System.out.print("Exact Match! You won $10,000");
		}
		else if(Math.ab(lotteryNum - guess) <= 4)
		{
			System.out.print("Within 4. You won $3,000");
		}
		else if(Math.ab(lotteryNum - guess) <= 8)
		{
			System.out.print("Within 8. You won $1,000");
		}
		else {
			System.out.print("You did not win. ");
		}
	}
}
```

7) Write a `MovieTicket` class that includes:

- Instance variables for the movie title, time of the show (as an integer in 24-hour format), and the viewer's age
- A default constructor that sets the title to "Unknown", the viewer's age to 12, and the time of show to 19 for 7 PM
- A parameter constructor that initializes two parameters to the viewer's age and time of the show. Add a selection statement inside the constructor to validate the showtime. If the showtime is not in the range 0-23, set it to the default value (19 for 7 PM)
- A method `calculatePrice()` that determines and returns the integer ticket price value based on the following rules:
    - Children (under 12): $8
    - Seniors (65 and older): $7
    - Regular price: $12
    - Discount for shows before 5 PM: $2 off all ticket types

Write the full class in the space below. There is also space on the next page to continue your class if you need more room.

```
public class MovieTicket {

    private String movieTitle;
    private int showTime;
    private int viewerAge;

    public MovieTicket() {
        movieTitle = "Unknown";
        viewerAge = 12;
        showTime = 19; // Default to 7 PM
    }

    public MovieTicket(String title, int age, int time) {
        movieTitle = title;
        viewerAge = age;

        if (time >= 0 && time <= 23) {
            showTime = time;
        }
        else {
            showTime = 19;
        }
    }

    public int calculatePrice() {
        int price;

        if (viewerAge < 12) {
            price = 8;
        }
        else if (viewerAge >= 65) {
            price = 7;
        }
```

```
        else {
            price = 12;
        }

        // Apply discount
        if (showTime < 17) {
            price -= 2;
        }

        return price;
    }
}
```

**D** 8) Consider the following code segment.

```
int x = 3;
int y = -1;

if(x - 2 > y)
{
      x -= y;
      if(x * 2 < 3)
      {
            x--;
      }
}
if(y + 3 >= x)
{
      y += x;
}
else
{
      y -= x;
}
```

x = 3, y = –1 → 3 – 2 is greater than –1

x = 4, y = –1 → 4 * 2 is not less than 3

x = 4, y = –1 → –1 + 3 is not greater than or equal to 4

x = 4, y = –5

What is the value of x and y after the code segment has been evaluated?

(A) x is 3 and y is –4
(B) x is 4 and y is –4
(C) x is 3 and y is –5
(D) x is 4 and y is –5

**C** 9) Consider the following expression. Assume that x and y are properly initialized `boolean` variables.

```
(x && y) || !(x || y)
```

The result of evaluating the expression is best described as:

(A) always true
(B) true only when x is true and y is true
(C) true only when x and y have the same values
(D) true only when x and y have different values

| x | y | x && y | !(x || y) | expression |
|---|---|---|---|---|
| T | T | T | F | T |
| T | F | F | F | F |
| F | T | F | F | F |
| F | F | F | T | T |

Now that we have the results of the expression, see which statement makes sense.

**B** 10) Consider the following code:

```
if(!(val >= 80 && val < 90))
```

Which of the following code fragments would have the same effect?

(A) `if(val >= 80 || val < 90)`
(B) `if(val < 80 || val >= 90)`
(C) `if(val < 80 && val >= 90)`
(D) `if(val <= 80 && val > 90)`

**B** 11) Which of the following statements is shown in the final column of the truth table?

| A | B | ? |
|---|---|---|
| T | T | F |
| T | F | F |
| F | T | T |
| F | F | F |

(A) `!(!A || B)`
(B) `!(A || !B)`
(C) `!(A && !B)`
(D) `!(!A && B)`

B 12) Which of the following will correctly display “Elementary” if the grade is 5 or lower, “Middle” if the grade is between 6 and 8, and “High” if the grade is 9 or above?

(A)

```
if(grade <= 5)
    System.out.print("Elementary");
if(grade <= 8)
    System.out.print("Middle");
if(grade >= 9)
    System.out.print("High");
```

(B)

```
if(grade <= 5)
    System.out.print("Elementary");
else
    if(grade <= 8)
        System.out.print("Middle");
    else
        System.out.print("High");
```

(C)

```
if(grade >= 9)
    System.out.print("High");
else
    if(grade <= 8)
        System.out.print("Middle");
    else
        System.out.print("Elementary");
```

(D)

```
if(grade <= 5)
    System.out.print("Elementary");
if(grade <= 8)
    System.out.print("Middle");
else
    System.out.print("High");
```

C 13) Consider the following code segment.

```
int myNumber = 5;
boolean myFlag = true;

if(myFlag)   //TRUE
{
     myNumber++;
}
if(myFlag || false)   //TRUE
{
     myNumber++;
     myFlag = false;
}
if(myFlag && false) //FALSE
{
     myNumber++;
}
```

| myNumber | myFlag |
|---|---|
| 5 | true |
| 6 | |
| 7 | false |

What is the value of `myNumber` after the above code executes?

(A) 5
(B) 6
(C) 7
(D) 8

B 14) Consider the code segment below.

```
int a = 4;
int b = 10;

if(a < 5) {   //TRUE
    if(b > 8) { //TRUE
        System.out.print("Path A ");
    }
    if (b > 5) { //TRUE
        System.out.print("Path B ");
    }
    else { //Skipped
        System.out.print("Path C ");
    }
}
else { //Skipped
    System.out.print("Path D");
}
```

What is printed when the code segment executes?

(A) Path A (B) Path A Path B (C) Path B Path D (D) Path D